

UTTAM TIWARI

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SUMMARY

Computer Science & AI student with a strong foundation in Python, Generative AI, and production-grade application development. Experienced in building LLM-based systems, multi-agent pipelines, RAG architectures, and automation solutions. Proven ability to collaborate across teams, deliver scalable software, and solve real-world problems using modern AI and cloud frameworks. Passionate about autonomous AI agents, process optimization, and engineering innovation.

EDUCATION

Medi-Caps University

B.Tech in Computer Science & Artificial Intelligence

Indore, MP

2023 – 2027

EXPERIENCE

Machine Learning Intern

Robotronix Tech Pvt. Ltd.

Jun 2025 – Sep 2025

Indore, MP

- Improved fraud detection model accuracy from 78% to 87% through feature engineering and hyperparameter optimization using Scikit-learn pipelines, contributing to production-ready performance.
- Built and evaluated deep learning pipelines for real-world classification tasks; analyzed model health via precision-recall curves and confusion matrix analysis.
- Documented pipeline changes and communicated findings effectively with cross-functional team members, improving technical communication practices.

PROJECTS

Misinfo Buster | WhatsApp Misinformation Fact-Checker | Python, LangGraph, RAG, Streamlit

🌐 GitHub

- Architected a multi-agent reasoning pipeline using LangGraph to autonomously verify forwarded WhatsApp messages, demonstrating end-to-end application development on cloud-deployable infrastructure.
- Designed a 3-way classification system (True / False / Uncertain) with structured prompt engineering; wired 3 modular LLM API calls for claim decomposition, retrieval, and response synthesis.
- Achieved ~2–3s end-to-end latency via optimized LangGraph orchestration with RAG-based semantic search; deployed live on Streamlit with iterative prompt tuning and test-query evaluation.

YouTube Helper Chatbot | Video RAG Assistant | Python, Gemini API, Vector DB

- Built a production-grade RAG pipeline in Python enabling context-aware Q&A over YouTube transcripts using natural-language-to-knowledge retrieval.
- Engineered a vector search pipeline with semantic chunking (400/100 overlap) and top-6 context retrieval; integrated Google Gemini API for reasoning and structured response generation.

TECHNICAL SKILLS

Languages: Python, Java, SQL

AI & GenAI: LLM APIs, Prompt Engineering, RAG, Hybrid RAG, Multi-Agent Systems, LangChain, LangGraph, NLP, Vector Databases, Agentic AI, Tool Calling, MCP

Cloud & DevOps: AWS, GCP, Docker, Terraform

Web & Frameworks: REST APIs, Streamlit

Version Control: Git, GitHub

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS & CERTIFICATIONS

Team Lead — Smart India Hackathon 2024

Internal Round, Runner-Up

Led a 6-member team to build an ML-based railway anomaly detection system

- Designed a delay-pattern prioritization model; demonstrated cross-functional leadership and collaborative problem-solving under competition constraints.

BCG X GenAI Job Simulation

Forage

AI chatbot development & financial document analysis using LLMs

IBM Applied AI Certification

IBM

Fundamentals of AI, Watson, and practical AI application development